

Week 5 — Classic IP Models, Networks, and the LAP

OL7014 Integer Programming • Instructor: Prof. H. Park • Week of Sep 21, 2026

1. The LP relaxation — and the question it leaves

LP relaxation: delete the integrality line — optimize over P instead of $X = P \cap \mathbb{Z}^n$

$$(\text{IP}) \quad \max_x \{ c^\top x : Ax \leq b, x \in \mathbb{Z}_+^n \} \quad \longrightarrow \quad (\text{LP}) \quad \max_x \{ c^\top x : Ax \leq b, x \geq 0 \}$$

larger feasible set $\Rightarrow$ always optimistic: $z_{\text{LP}} \geq z_{\text{IP}}$ (max), $z_{\text{LP}} \leq z_{\text{IP}}$ (min)

week 2: $22.5 \geq 19$ on the toy, rounding worth only 18 — a **bound**, not an answer. bounds solve every IP (week 7)

today: when is the gap zero?

gap = 0 $\Rightarrow$ write $x \in \mathbb{Z}$, solve the relaxation, get an integer plan anyway — *the LP is the IP*

not luck, not rounding: a whole family (assignment, transportation, networks), readable off A

KEY one line, $\in \mathbb{Z}$, is what makes an IP hard — so a model where deleting it costs nothing is worth recognizing on sight.

2. Assignment — a model born canonical

demo ip-games / assignment — play FIRST: four cadets, four branches, one each — the board below is the same model at $n = 3$. <https://hankpark0706.github.io/ip-games/assignment-matching.html>

3 pilots, 3 aircraft, one each — officers to billets, units to sectors: the **assignment problem** (할당 문제)

c_{ij} = value of pairing (i, j) ; the decision is a *table*: $X \in \{0, 1\}^{3 \times 3}$, $x_{ij} = 1$ iff pilot i flies aircraft j

rows sum to 1 (each pilot once), columns sum to 1 (each aircraft once) — week 2's BP:

$$\begin{aligned} \max_X \quad & \langle C, X \rangle \\ \text{s.t.} \quad & X\mathbf{1} = \mathbf{1}, \quad X^\top \mathbf{1} = \mathbf{1} \\ & x_{ij} \in \{0, 1\} \end{aligned}$$

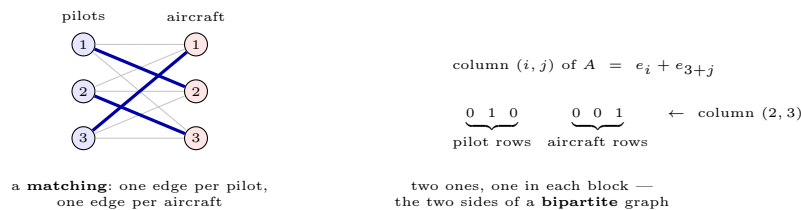
read the symbols: $\langle C, X \rangle = \sum_{i,j} c_{ij} x_{ij}$, $\mathbf{1}$ = all-ones vector

flatten: $x = \text{vec}(X) \in \mathbb{R}^9 \Rightarrow A$ is 6×9 , all 0/1, $b = \mathbf{1}$

- each column: exactly **two ones** — one pilot row, one aircraft row (an *incidence matrix*)

- rank $A = 5$: any one row is implied by the rest

relax (section 1): $x_{ij} \in \{0, 1\} \rightarrow x_{ij} \geq 0$ (row sums already cap at 1) $\Rightarrow$ the LP returns a **matching**, every cost table, every time



KEY $2n$ rows, n^2 binaries, harder-looking than week 1's two-row toy — and the gap is zero.

takeaway delete assignment's integrality line and the answer does not move. the reason has a name — *total unimodularity* — coming up.

3. Transportation — assignment with quantities

supply s_i at source i , demand d_j at sink j (assignment: all = 1) $\Rightarrow$ matching becomes **flow**

x_{ij} = amount shipped $i \rightarrow j$ — crates, pallets, sorties: whole units, so integrality is declared

$$\begin{aligned} \min_x \quad & \sum_{i=1}^n \sum_{j=1}^k c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^k x_{ij} \leq s_i \quad (i \in [n]) \quad \text{ship no more than you hold (공급 한도)} \\ & \sum_{i=1}^n x_{ij} \geq d_j \quad (j \in [k]) \quad \text{every sink gets what it needs (수요 충족)} \\ & x_{ij} \in \mathbb{Z}_+ \end{aligned}$$

balanced: $\sum_i s_i = \sum_j d_j$ with $c \geq 0$ nobody ships spare, so the $\geq d_j$ rows may be written =

relax: $x_{ij} \in \mathbb{Z}_+ \rightarrow x_{ij} \geq 0 \Rightarrow$ integer s, d give whole shipments back. gap zero again — and not a binary in sight

assignment is the special case $k = n$, $s_i = d_j = 1$: row sums cap x_{ij} at 1, and section 4 puts the optimum on a *corner* — a permutation matrix

takeaway **transportation = assignment with quantities: supplies left, demands right, flows between. an IP you may hand to an LP solver whenever the data is whole.**

4. Total unimodularity — why the LP told the truth

gap on week 1's toy: 22.5 – 19. gap on assignment, transportation: 0. same solver $\Rightarrow$ the difference is in the *matrix*

Def. A is **totally unimodular (TU)** (완전 유니모듈러 행렬): every square submatrix has determinant 0, +1, or -1 (any k rows $\times$ any k columns, every k) — a property of A alone: no b , no c

$k = 1$ is **free**: a 1×1 submatrix is one entry $\Rightarrow$ **every entry of a TU matrix is $0, \pm 1$**

week 2's rucksack row (5, 7, 8, 8, 2) has an 8 $\Rightarrow$ not TU (the numbers, not the shape: $\sum_i x_i \leq p$ is TU)

Prop. (Hoffman–Kruskal) A TU **and** b integer $\Rightarrow$ every corner of $P = \{x : Ax \leq b, x \geq 0\}$ is integer (same for $Ax = b$)

$\Rightarrow$ the LP, if it has an optimum, has an *integer* one: **the relaxation solves the IP**

mechanism: in $Ax = b, x \geq 0$ a corner picks *basic columns* B (the rest are 0), and $A_B x_B = b$. Cramer:

$$(x_B)_j = \frac{\det(A_B \text{ with column } j \text{ replaced by } b)}{\det A_B}$$

integer on top; $\det A_B = \pm 1 \Rightarrow$ no fraction can appear. ($\det = 0$: not a basis, never a corner)

KEY **TU is the promise that *no* corner can be fractional, whatever integer b you throw at it.**

5. Recognizing it — two dimensions, then the checklist

week 1's objective $\max 4x + 5y, x, y \geq 0$, change *nothing but* A :

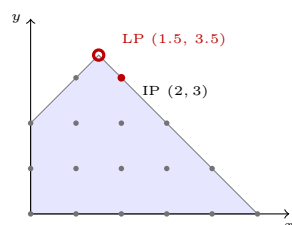
$$(P_0) \begin{cases} x + 3y \leq 10 \\ 3x + y \leq 10 \end{cases} \quad (P_1) \begin{cases} x + y \leq 5 \\ -x + y \leq 2 \end{cases} \quad (P_2) \begin{cases} x + y \leq 5 \\ y \leq 2 \end{cases}$$

(P_0) = week 1's toy. (P_1) : two teams fly ≤ 5 sorties between them, B at most two *more than* A. (P_2) : flat cap, B flies ≤ 2

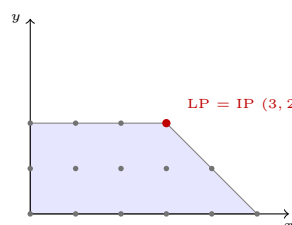
	A	largest $ \det $	TU?	LP optimum	IP optimum
(P_0)	$\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$	8	no — there is a 3	(2.5, 2.5), 22.5	(1, 3), 19
(P_1)	$\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$	2	no — $\det = 2$	(1.5, 3.5), 23.5	(2, 3), 23
(P_2)	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	1	yes	(3, 2), 22	(3, 2), 22

(adding the rows $-x \leq 0, -y \leq 0$ changes nothing here)

(P_1) vs (P_2) : **one entry** apart. $\det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = 2 \Rightarrow$ a division by 2 $\Rightarrow$ the 1.5. $-1 \rightarrow 0$: $\det = 1$, corners snap to integers



(P_1) : $\det = 2$, **not** TU
the top corner is fractional



(P_2) : $\det = 1$, **TU**
every corner is a lattice point

two traps. 1. entries all $0, \pm 1 \not\Rightarrow$ TU — see (P_1) . 2. TU belongs to the *whole* matrix: neither row of (P_1) is guilty alone

checklist (sufficient, not necessary) — nobody checks determinants:

- (i) every entry is $0, \pm 1$
- (ii) every column has at most two nonzeros
- (iii) rows split into M_1, M_2 so that **in every column with two nonzeros** the M_1 entries and the M_2 entries sum to the same thing

two ways to pass (iii) — the whole course:

- **two** +1s (*undirected* incidence): opposite sides $\Rightarrow M_1, M_2 =$ a **bipartition**. pilots / aircraft **are** M_1 / M_2
- **one** +1, **one** -1 (*directed* incidence): $M_1 =$ all rows, $M_2 = \emptyset$ — any graph (section 6)

model	matrix	LP relaxation
week 1's toy (P_0)	a row with a 3 in it	fractional corner (2.5, 2.5)
assignment	bipartite incidence, TU	integral: the LP <i>is</i> the IP
transportation	bipartite incidence, TU	integral whenever s, d are

demo `demo1.toy.ip.py` — week 2's demo on week 1's toy: r_1, r_2 move b — no slider moves A , and A is where TU lives. at $r_1 = r_2 = 4$ the LP does land on (1, 1): failing TU means *some* integer b gives fractions, not every one.

takeaway first time in this course that the *shape* of A — not its size, not the solver — decided whether a problem was easy.

6. Minimum cost network flow (MCNF) — the family worth recognizing

demo `ip-games / min-cost-flow` — push flow through a network by hand, then let the solver do it. <https://hankpark0706.github.io/ip-games/min-cost-network-flow.html>

directed graph $G = (V, E)$, $|V| = m$ nodes (노드), arcs (아크) $(i, j) \in E$

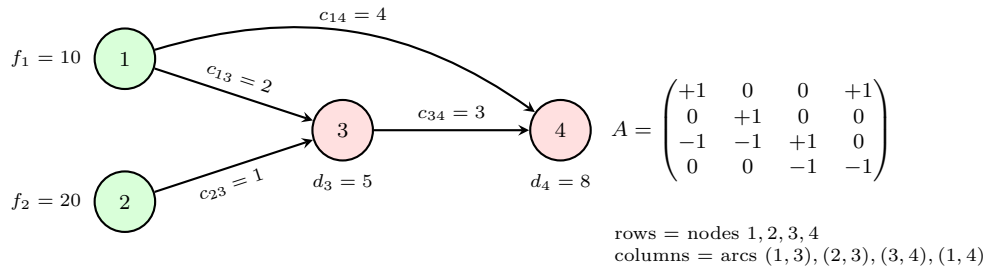
node i : $b_i =$ net supply = outflow - inflow **arc** (i, j) : cost c_{ij} , bounds $l_{ij} \leq x_{ij} \leq u_{ij}$ **decision**: $x_{ij} =$ flow (whole units $\Rightarrow$ integer)

$b_i > 0$	node i supplies	source (공급지)
$b_i < 0$	node i demands	destination (수요지)
$b_i = 0$	flow only passes through	transshipment , a hub (환적지)

$$\begin{aligned} \min_x \quad & \sum_{(i,j) \in E} c_{ij} x_{ij} \\ \text{s.t.} \quad & \underbrace{\sum_{j: (i,j) \in E} x_{ij} - \sum_{j: (j,i) \in E} x_{ji}}_{\text{one row per node — flow conservation (흐름 보존)}} = b_i \quad (\forall i \in V) \\ & l_{ij} \leq x_{ij} \leq u_{ij}, \quad x_{ij} \in \mathbb{Z}_+ \end{aligned}$$

rows sum to zero $\Rightarrow \sum_i b_i = 0$ required — else add a zero-cost **dummy node**

node-arc incidence matrix: one row per node, one column per arc, +1 where the arc leaves, -1 where it arrives



checklist: (i) entries $0, \pm 1$ ✓ (ii) two per column ✓ (iii) $M_1 = V, M_2 = \emptyset$ ✓ — *every* directed graph passes (undirected: two +1s need a bipartite split — the **matching LP** is integral only on bipartite graphs)

rank $A = |V| - 1 = 3$: one conservation row is always redundant

Prop. integer $\{b_i\}$ and integer bounds $\{l_{ij}\}, \{u_{ij}\} \Rightarrow$ every extreme point is integral: delete $x \in \mathbb{Z}_+$, solve the LP, **no branching, ever**

one model, four classics:

problem	is MCNF with
transportation	sources $b_i = s_i > 0$, destinations $b_j = -d_j$
assignment	transportation with every $b_i = +1, b_j = -1$ (equally many of each)
shortest path	$b_s = +1, b_t = -1, b_i = 0$ otherwise
max flow	all $b_i = 0$; add an artificial arc (t, s) and maximize x_{ts}

sections 2 and 3 were the same model twice — and both were networks

the picture is not yet an instance: nodes 3, 4 demand ($b_3 = -5, b_4 = -8$), so nodes 1, 2 must supply 13 — *which one, and how much, is not data*. that is the LAP ($f_1, f_2 =$ opening costs, section 7)

KEY recognize an IP as a network and it is solved: one row per node, one column per arc, ± 1 entries — TU by construction, so the relaxation is free.

7. Facility location — paying to open

transportation assumed the depots existed — *which depots should exist?* (시설 입지 문제)

watch the letters: back to week 2's convention — x = integer (open or not), y = continuous. relief tons are divisible, so the flow is no longer the integer decision

$x_i = 1$ iff depot i opens, f_i = fixed opening cost, y_{ij} = tons shipped

$$\begin{aligned} \min_{x, y} \quad & \sum_i f_i x_i + \sum_i \sum_j c_{ij} y_{ij} \quad \text{fixed + shipping} \\ \text{s.t.} \quad & \sum_i y_{ij} = d_j \quad (\text{demand}) \\ & y_{ij} \leq M x_i \quad (\text{linking — week 2}) \\ & y_{ij} \geq 0, \quad x_i \in \{0, 1\} \end{aligned}$$

$x_i = 0 \Rightarrow y_{ij} = 0$; $x_i = 1$ releases it — *uncapacitated*

how big is M ? $y_{ij} \leq d_j$ always $\Rightarrow M = d_j$, smallest valid. $M = \sum_j d_j$: valid, but on section 6's network the LP bound drops 52 $\rightarrow$ 47

1. **fix $x \Rightarrow$** transportation falls out: *all the hardness is in the binaries*

2. **big- M costs the guarantee:** $y_{ij} \leq M x_i$ belongs to no incidence matrix $\Rightarrow$ MCNF's proposition is gone. (not TU $\neq$ the LP lies: only *some* b must go fractional. section 8 checks)

KEY every guarantee in this note is a property of A — so every one is a single modeling decision away from being lost.

8. The LAP — when b is a decision

MCNF takes b_i as data; the **location-allocation problem** (입지-배분 문제) chooses it

$r_i \geq 0$ = reserve held at i (cost h_i /ton), $q_i \geq 0$ = unmet demand (penalty u_i), $s_i \geq 0$ = surplus (penalty o_i ; not section 3's supply)

$$b_i = \underbrace{r_i}_{\text{decision}} - d_i + q_i - s_i \quad \Rightarrow \quad \underbrace{\sum_j y_{ji} - \sum_j y_{ij}}_{\text{inflow} - \text{outflow}} + r_i + q_i - s_i = d_i \quad (\forall i \in V)$$

$$\begin{aligned} \min_{x, r, y, q, s} \quad & \sum_i f_i x_i + \sum_i h_i r_i + \sum_{(i,j) \in E} c_{ij} y_{ij} + \sum_i u_i q_i + \sum_i o_i s_i \\ \text{s.t.} \quad & \sum_j y_{ji} - \sum_j y_{ij} + r_i + q_i - s_i = d_i \quad (\forall i) \quad [\text{conservation — MCNF}] \\ & \sum_{i \in S} r_i \leq B \quad [\text{budget — a knapsack row}] \\ & r_i \leq B x_i \quad (\forall i \in S) \quad [\text{linking — week 2}] \\ & \sum_{i \in S} x_i \leq p \quad [\text{at most } p \text{ depots — week 2}] \\ & y, r, s \geq 0, \quad 0 \leq q_i \leq d_i, \quad x_i \in \{0, 1\} \end{aligned}$$

five costs: open (f_i), hold (h_i), ship (c_{ij}), fail (u_i), waste (o_i). every constraint is old

$S \subseteq V$ = **candidate sites**; elsewhere $x_i = r_i = 0$ — no reserve inside the disaster area

$M = B$ passes, since the budget row already caps each r_i at B — valid, though week 2 would ask for the *smallest* valid M

section 6's network, priced — $S = \{1, 2\}$, $B = 15$, $p = 1$, $h_i = 1$, $u_i = 100$, $o_i = 1$:

$$\begin{aligned} \min \quad & 10x_1 + 20x_2 + r_1 + r_2 + 2y_{13} + y_{23} + 3y_{34} + 4y_{14} + 100(q_3 + q_4) + \sum_i s_i \\ \text{s.t.} \quad & -y_{13} - y_{14} + r_1 + q_1 - s_1 = 0 \quad [\text{node 1}] \\ & -y_{23} + r_2 + q_2 - s_2 = 0 \quad [\text{node 2}] \\ & y_{13} + y_{23} - y_{34} + q_3 - s_3 = 5 \quad [\text{node 3}] \\ & y_{34} + y_{14} + q_4 - s_4 = 8 \quad [\text{node 4}] \\ & r_1 + r_2 \leq 15, \quad r_1 \leq 15x_1, \quad r_2 \leq 15x_2, \quad x_1 + x_2 \leq 1 \\ & y, r, s \geq 0, \quad q_1 = q_2 = 0, \quad 0 \leq q_3 \leq 5, \quad 0 \leq q_4 \leq 8, \quad x_1, x_2 \in \{0, 1\} \end{aligned}$$

sixteen variables, two binary ($q_1 = q_2 = 0$ since $d_1 = d_2 = 0$, so fourteen move). at $p = 1$, two plans:

plan	open	hold	ship	total
$x_2 = 1$: $r_2 = 13$, $y_{23} = 13$, $y_{34} = 8$	20	13	$1 \cdot 13 + 3 \cdot 8 = 37$	70
$x_1 = 1$: $r_1 = 13$, $y_{13} = 5$, $y_{14} = 8$	10	13	$2 \cdot 5 + 4 \cdot 8 = 42$	65

site 2 ships cheaper and still loses: -5 on the road, $+10$ at the door

$f_1 = 15 \Rightarrow$ tie at 70; past that, or arc (1, 4) flooded (site 1: $10 + 13 + 50 = 73$), site 2 wins

a cost paid once against a cost paid per ton — that trade is the LAP

and the LP lies, as section 7 warned:

$$z_{\text{LP}} = 62 < z_{\text{IP}} = 65, \quad x_1 = \frac{8}{15}, \quad x_2 = \frac{1}{3}, \quad r_1 = 8, \quad r_2 = 5$$

half-open depots: open for $10 \cdot \frac{8}{15} + 20 \cdot \frac{1}{3} = 12$, hold 13, ship 37 — site 2's road bill without site 2's door $\Rightarrow 62$.
section 1's gap, on our own toy

demo `lap_app.py` — the model on a map — drag **training data** to 1 and the dashboard collapses to today's single-scenario model (above 1 it is week 9's). **max facilities** is p ; prices are **budget** $B = B$, **facility cost** $c = f_i$, **reserve cost** $f = h_i$ — the app's letters do not match ours. *Solve* picks the plan, *Simulate disaster* routes it.

!takeaway **LAP = MCNF + two decisions MCNF cannot make: *which nodes supply* (x_i) and *how much* (r_i).**
fix those and a flow problem is all that is left — TU, easy. all the hardness is in (x, r).

9. The assumption we just made

d_i on the right-hand side, *known* — when x_i, r_i are chosen. nobody knows where the earthquake will be

stage 1: now	$\longrightarrow$ <i>the disaster</i> $\longrightarrow$	stage 2: then
open depots, place stock	demand d reveals itself	route what you have
x_i, r_i		y_{ij}, q_i, s_i
committed, binary, expensive		reactive, a flow problem, easy

the split is already *in the variables* — re-read the model, do not rebuild it. worst case or average? — week 9

KEY *ML predicts; optimization decides* — **a forecast of d is an input, not an answer.**

next lecture is $r_i \leq Bx_i$ the *best* way to write that row? other formulations describe the same plans and return the same answer — and one solves in seconds where another runs all night. **Homework 2 is out** (due 10/6).